

Операционные системы

Продвинутое использование git

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
ramielsarhanov@ramielsarhanov:~$ pnpm setup
Appended new lines to /home/ramielsarhanov/.bashrc

Next configuration changes were made:
export Pnpm_HOME="/home/ramielsarhanov/.local/share/pnpm"
case ":$PATH:" in
  *:$Pnpm_HOME:*) ;;
  *) export PATH="$Pnpm_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/ramielsarhanov/.bashrc
ramielsarhanov@ramielsarhanov:~$ source ~/.bashrc
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 1: Node.js

```
ramielsarhanov@ramielsarhanov:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/ramielsarhanov/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
Done in 2.7s using pnpm v10.27.0
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 2: установка commitizen


```
ramielsarhanov@ramielsarhanov:~$ pnpm add -g standard-changelog
WARN GET https://registry.npmjs.org/conventional-changelog error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29223ms: https://registry.npmjs.org/conventional-changelog
WARN GET https://registry.npmjs.org/@simple-lib%2Fstream-utils error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29222ms: https://registry.npmjs.org/@simple-lib%2Fstream-utils
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/ramielsarhanov/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 59.1s using pnpm v10.27.0
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 3: установка standard-changelog

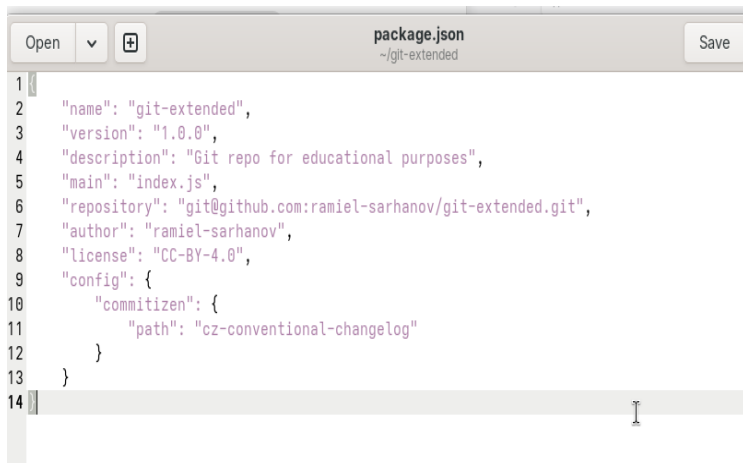


Рисунок 4: package.json

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git add .
ramielsarhanov@ramielsarhanov:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't modify
src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master cf40bd4] chore: test
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
ramielsarhanov@ramielsarhanov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
 19ea55f..cf40bd4  master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 5: Отправка

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/ramielsarhanov/git-extended/.git/hooks]
ramielsarhanov@ramielsarhanov:~/git-extended$ git branch
* develop
  master

ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/ramiel-sarhanov/git-extended/pull/new/develop
remote:
To github.com:ramiel-sarhanov/git-extended.git
* [new branch]      develop -> develop
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

ramielsarhanov@ramielsarhanov:~/git-extended$ standard-changelog --first-release
ramielsarhanov@ramielsarhanov:~/git-extended$ git add CHANGELOG.md
ramielsarhanov@ramielsarhanov:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 b201f78] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
Already on 'master'
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.68 KiB | 2.68 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
   cf40bd4..eb20261  develop -> develop
   cf40bd4..c80aeba  master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/git-extended.git
 * [new tag]         1.0.0 -> 1.0.0
ramielsarhanov@ramielsarhanov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/ramiel-sarhanov/git-extended/releases/tag/v1.0.0
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 8: Отправка

```
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

ramielsarhanov@ramielsarhanov:~/git-extended$ touch test2
ramielsarhanov@ramielsarhanov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was eb20261).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.79 KiB | 2.79 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/git-extended.git
   eb20261..345bbe5  develop -> develop
   c80aeba..a2fec6f  master -> master
ramielsarhanov@ramielsarhanov:~/git-extended$
```

Рисунок 10: Завершение релиза

Подготовка рабочего репозитория

```
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git add .
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't modify
src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master dbd347b] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
   b5ecd08..dbd347b  master -> master
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$
```

Рисунок 11: package.json и коммит

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.77 KiB | 2.77 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
   dbd347b..c0e8d19  develop -> develop
   dbd347b..36242d6  master -> master
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ramiel-sarhanov/2026-1--study--os-intro.git
 * [new tag]           1.0.0 -> 1.0.0
ramielsarhanov@ramielsarhanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
o$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.